Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: DSIP (01CT1513)	**Aim: To design digital filters and study their effects on signals and MP3 audio.**	
Open-ended-Problem: 04	**Date:22-07-2026**	**Enrollment No:92400133121**

Aim: To design digital filters and study their effects on signals and MP3 audio

(a)
Select any MP3 song of your choice and process it using a low-pass filter that you design.

Experiment with at least three different cutoff frequencies and observe how each setting changes the filtered audio output.

Discuss your observations in detail, focusing on how the cutoff frequency influences the clarity, tonal balance, and overall character of the song.

(b) Use an appropriate band-pass filter to separately suppress vocals and background music, saving each filtered result as its own MP3 file.

Analyze and describe your observations, focusing on how each filter affects the different components of the song.

Theory:

- Digital filters are used to process signals by allowing certain frequency components to pass while reducing others.
- They are widely used in audio processing to improve sound quality and control the frequency content of a signal.

- Butterworth Filter:
  - A Butterworth filter has a smooth and flat passband response.
  - It is used to remove unwanted frequency components while maintaining a smooth frequency response.

- Chebyshev Filter:
  - A Chebyshev filter provides a sharper transition between the passband and stopband than a Butterworth filter of the same order.
  - However, a Type I Chebyshev filter has ripples in its passband.

- Bilinear Transformation:
  - Bilinear transformation is a method used to convert an analog filter into a digital filter.
  - It maps the analog filter's frequency response into the digital domain.

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- Low-Pass Filter:
  - A low-pass filter allows frequencies below a specified cutoff frequency to pass and attenuates higher frequencies.
  - In audio processing, it can reduce high-frequency sounds and make audio sound less sharp.

- Band-Pass Filter:
  - A band-pass filter allows frequencies within a selected range to pass while attenuating frequencies outside that range.
  - It can emphasize vocal-range or high-frequency instrumental components, but it cannot completely separate vocals from background music because their frequency ranges overlap.

Code:

```
import os
import numpy as np
import matplotlib.pyplot as plt
import soundfile as sf

from pydub import AudioSegment
from scipy.signal import butter, sosfiltfilt

input_file = "audio1.mp3"
output_folder = "output"

os.makedirs(output_folder, exist_ok=True)

audio = (
    AudioSegment.from_mp3(input_file)
    .set_sample_width(2)
    .set_frame_rate(44100)
)

sampling_frequency = audio.frame_rate
channels = audio.channels

samples = np.array(audio.get_array_of_samples(), dtype=np.float64)
samples = samples.reshape(-1, channels)

original_signal = samples / 32768.0
```

```
print("Song loaded successfully!")
print("Sampling frequency:", sampling_frequency, "Hz")
print("Number of channels:", channels)
print("Duration:", len(original_signal) / sampling_frequency, "seconds")
```

Output:

```
Song loaded successfully!
Sampling frequency: 44100 Hz
Number of channels: 2
Duration: 27.539977324263038 seconds
```

```
def save_mp3(signal, filename):

    peak = np.max(np.abs(signal))

    if peak > 0:
        signal = signal * (0.98 / peak)

    signal = np.clip(signal, -1, 1)

    pcm = (signal * 32767).astype(np.int16)

    audio_segment = AudioSegment(
        data=pcm.tobytes(),
        sample_width=2,
        frame_rate=sampling_frequency,
        channels=channels
    )

    path = os.path.join(output_folder, filename)

    audio_segment.export(
        path,
        format="mp3",
        bitrate="192k"
    )

    print("Saved:", path)

def plot_audio(signal, title):
```

```
    number_of_samples = min(
        int(0.05 * sampling_frequency), len(signal)
    )

    time = np.arange(number_of_samples) / sampling_frequency

    plt.figure(figsize=(10, 3))
    plt.plot(time, signal[:number_of_samples, 0])
    plt.title(title + " - Time Domain")
    plt.xlabel("Time (seconds)")
    plt.ylabel("Amplitude")
    plt.grid(True)
    plt.tight_layout()
    plt.show()

    spectrum = np.abs(np.fft.rfft(signal[:, 0]))
    frequency = np.fft.rfftfreq(
        len(signal), d=1 / sampling_frequency
    )

    magnitude_db = 20 * np.log10(
        spectrum / (np.max(spectrum) + 1e-12) + 1e-12
    )

    plt.figure(figsize=(10, 3))
    plt.plot(frequency, magnitude_db)
    plt.title(title + " - Frequency Domain")
    plt.xlabel("Frequency (Hz)")
    plt.ylabel("Magnitude (dB, relative)")
    plt.xlim(0, min(10000, sampling_frequency / 2))
    plt.ylim(-100, 5)
    plt.grid(True)
    plt.tight_layout()
    plt.show()


# Q3(a): BUTTERWORTH LOW-PASS FILTER

cutoff_frequencies = [500, 1500, 4000]

for cutoff in cutoff_frequencies:
```

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```
sos = butter(
    4,
    cutoff,
    btype="lowpass",
    fs=sampling_frequency,
    output="sos"
)

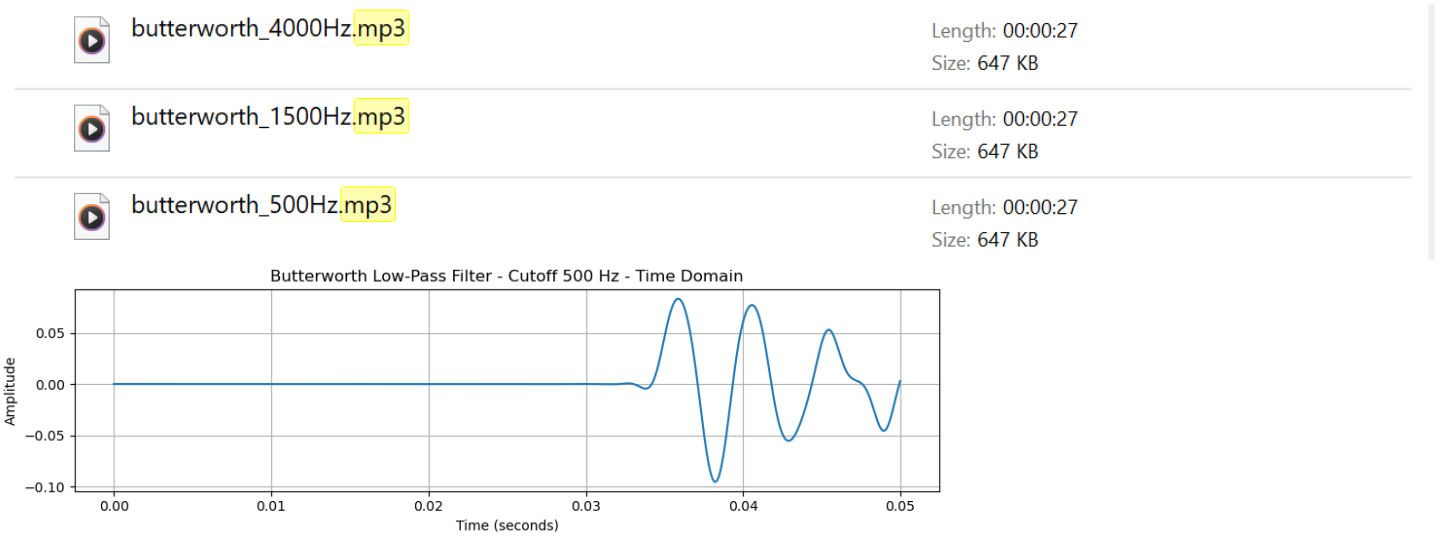
filtered_signal = sosfiltfilt(
    sos, original_signal, axis=0
)

filename = f"butterworth_{cutoff}Hz.mp3"

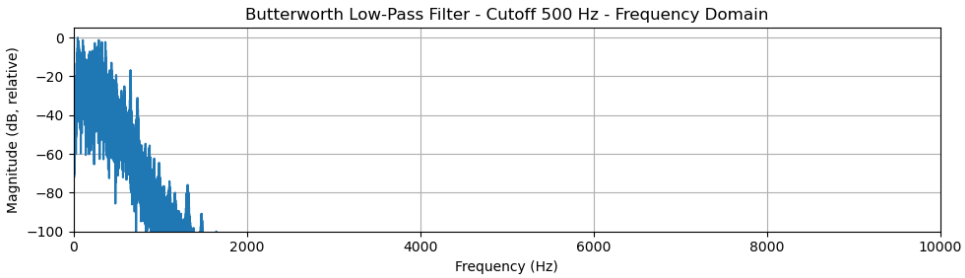
save_mp3(filtered_signal, filename)

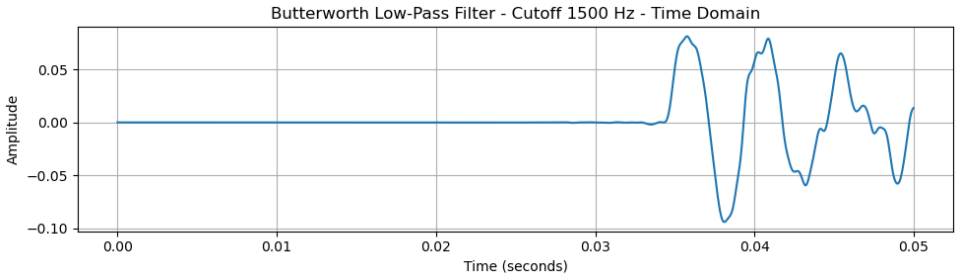
plot_audio(
    filtered_signal,
    f"Butterworth Low-Pass Filter - Cutoff {cutoff} Hz"
)
```

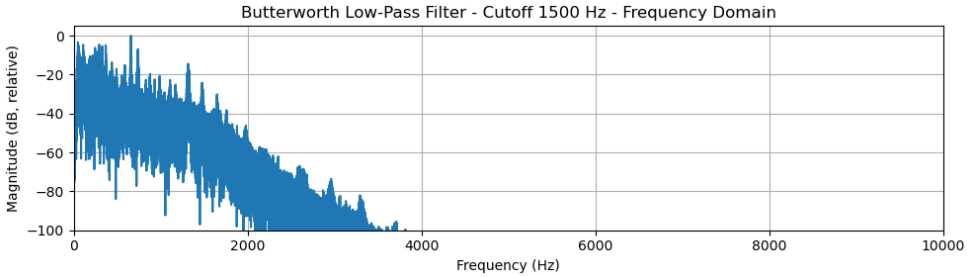
Output:

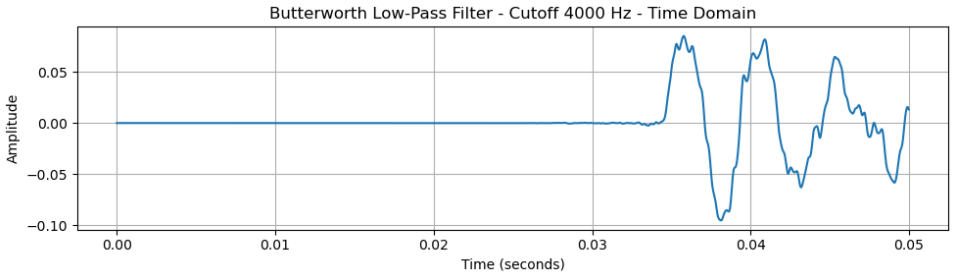


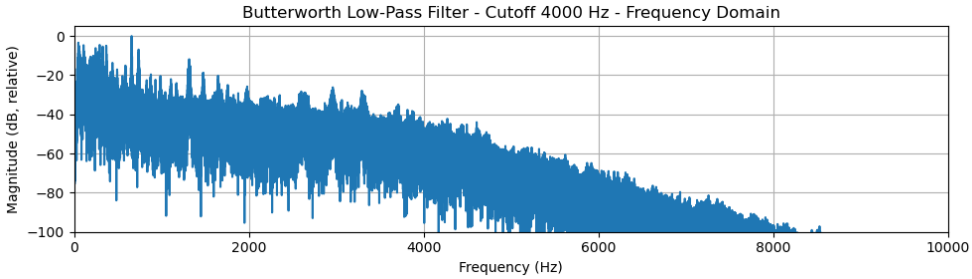
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Observation:

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- At a cutoff frequency of 500 Hz, high-frequency components were reduced, making the song more suppressed.
- At a cutoff frequency of 1500 Hz, more vocal characteristics were retained, but some high-frequency details and clarity were reduced.
- At a cutoff frequency of 4000 Hz, the song was clear, with more vocal and instrumental details preserved
- Increasing the cutoff frequency allows more high-frequency components to pass through the filter.

Conclusion:

- The audio song was processed using a Butterworth low-pass filter with cutoff frequencies of 500 Hz, 1500 Hz, and 4000 Hz.
- At lower cutoff frequencies, high-frequency components were attenuated.
- At higher cutoff frequencies, more vocal and instrumental details were preserved.
- Thus, changing the cutoff frequency affects the clarity and quality of the audio.

```
# Q3(b): VOCAL-DOMINANT BAND-PASS FILTER

vocal_low = 300
vocal_high = 3400

sos_vocal = butter(
    4,
    [vocal_low, vocal_high],
    btype="bandpass",
    fs=sampling_frequency,
    output="sos"
)

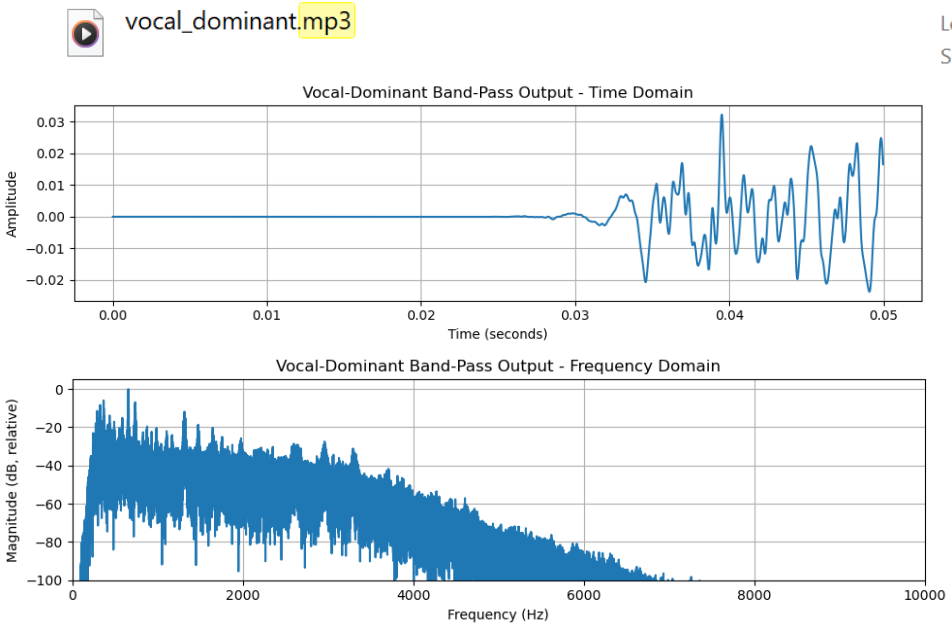
vocal_signal = sosfiltfilt(
    sos_vocal, original_signal, axis=0
)

save_mp3(vocal_signal, "vocal_dominant.mp3")

plot_audio(
    vocal_signal,
    "Vocal-Dominant Band-Pass Output"
)
```

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Output:



Observation

- The band-pass filter with a frequency range of 300–3400 Hz emphasized the frequency components associated with human speech and vocals.
- As a result, the vocals became more prominent in the filtered audio.
- However, some background instruments were also audible because their frequency components overlap with the vocal frequency range.
- Lower-frequency bass components and higher-frequency instrumental details were reduced to some extent.

Conclusion

- The band-pass filter successfully emphasized the vocal-dominant frequency range of the audio song.
- However, it could not completely separate the vocals from the background music because both contain overlapping frequency components.
- Thus, band-pass filtering can improve the prominence of vocals but cannot achieve complete vocal isolation.

```
# Q3(b): HIGH-FREQUENCY INSTRUMENTAL EMPHASIS

music_low = 4000
music_high = min(
    8000, sampling_frequency / 2 - 100
```

```
)

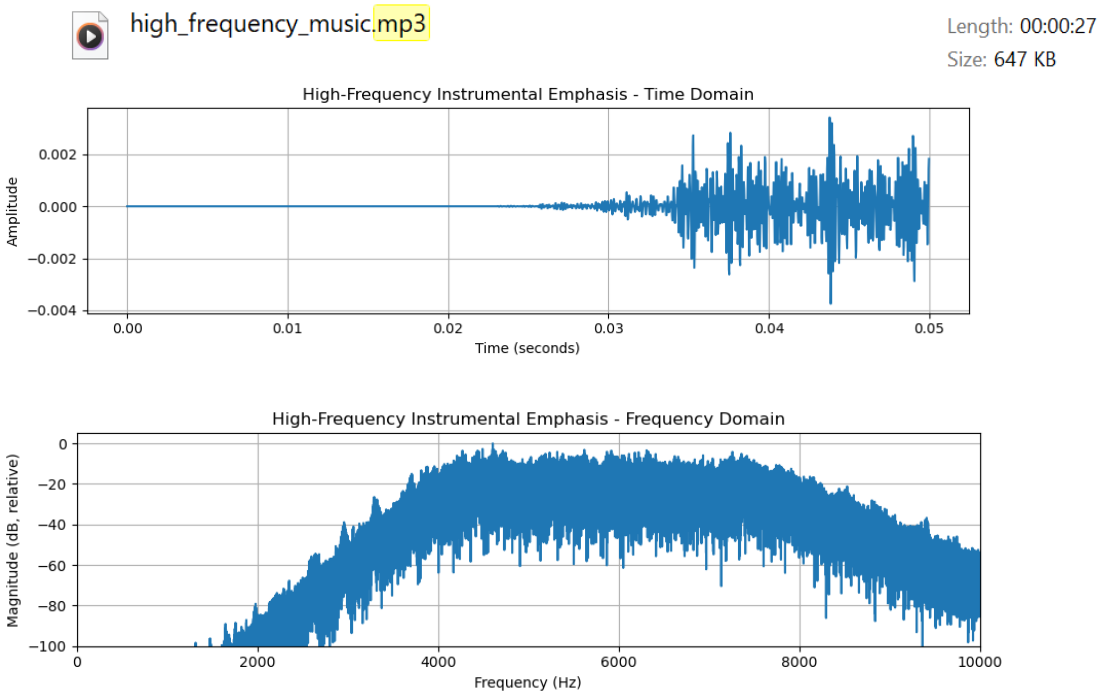
sos_music = butter(
    4,
    [music_low, music_high],
    btype="bandpass",
    fs=sampling_frequency,
    output="sos"
)

music_signal = sosfiltfilt(
    sos_music, original_signal, axis=0
)

save_mp3(music_signal, "high_frequency_music.mp3")

plot_audio(
    music_signal,
    "High-Frequency Instrumental Emphasis"
)
```

Output:



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Observation

- The band-pass filter with a frequency range of 4000–8000 Hz allowed high-frequency sounds to be heard more clearly.
- The bass and low-frequency sounds were reduced, and the vocals sounded less full compared to the original song.
- The filtered audio sounded thinner and brighter.
- However, some vocal sounds were still present in the output.

Conclusion

- The high-frequency band-pass filter helped to highlight high-frequency instrumental sounds while reducing bass and other low-frequency components.
- It changed the overall sound quality of the song, making it sound brighter.
- However, the background music could not be completely separated from the vocals because both contain sounds from overlapping frequency ranges.